

Unit 2.7 Vectors in Two Dimensions

Paper 1: Basic

- The point A is translated to $B(5, -2)$ by the vector $\begin{pmatrix} -3 \\ 7 \end{pmatrix}$.
 - Find the coordinates of point A . [1]
 - Find the magnitude of $\begin{pmatrix} -3 \\ 7 \end{pmatrix}$. [2]

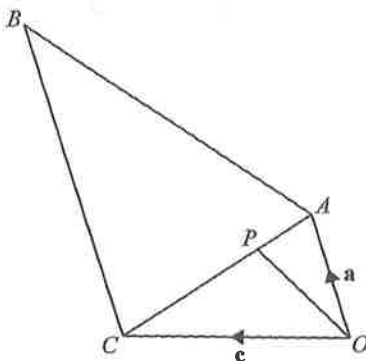
[N19/I/13]
- $\overrightarrow{AB} = \begin{pmatrix} -8 \\ 15 \end{pmatrix}$.
 - Find $|\overrightarrow{AB}|$. [1]
 - C is the point $(4, 20)$.
 $\overrightarrow{DC} = 2\overrightarrow{AB}$.
 Find the coordinates of D . [2]
 - What type of quadrilateral is $ABCD$? [1]

[N16/I/23]
- A is the point $(-1, 4)$ and $\overrightarrow{BA} = \begin{pmatrix} -3 \\ 6 \end{pmatrix}$.
 - Find the coordinates of point B . [1]
 - Calculate $|\overrightarrow{AB}|$. [1]

[N15/I/8]

Paper 1: Intermediate

- $OABC$ is a quadrilateral.
 $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$ and $\overrightarrow{OP} = m(\mathbf{a} - \mathbf{c})$.



- Write $\overrightarrow{OP}$ in terms of $\mathbf{a}$, $\mathbf{c}$ and m . [1]
- $\overrightarrow{OB} = 4(1 - m)\mathbf{c} + 4m\mathbf{a}$.
 - Show that O , P and B lie on a straight line. [2]
 - $\overrightarrow{CB} = k\mathbf{a}$.
 Find the value of m and the value of k . [3]
- Write down the ratio $CP : CA$. [1]

[N20/I/24]

5. (a) $\overrightarrow{PQ} = \begin{pmatrix} a \\ 3a \end{pmatrix}$.
 $|\overrightarrow{PQ}| = \frac{5\sqrt{10}}{2}$.

Find the two possible values of a .

[2]

(b) A line joins the two points $A(-5, 1)$ and $B(43, 38)$.

(i) Find $\overrightarrow{AB}$.

[1]

(ii) Use vectors to show whether or not the point $C(11, 13)$ lies on this line.

[2]

[N18/I/18]

6. ABC is a triangle.

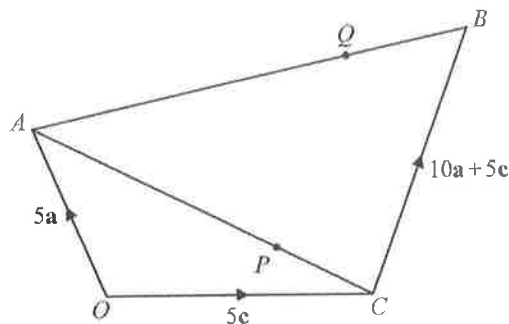
$$\overrightarrow{AB} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \overrightarrow{AC} = \begin{pmatrix} -1 \\ 8 \end{pmatrix}.$$

Calculate the length of BC .

[3]

[N17/I/12]

7.



$OABC$ is a quadrilateral.

$$\overrightarrow{OA} = 5\mathbf{a}, \overrightarrow{OC} = 5\mathbf{c}, \overrightarrow{CB} = 10\mathbf{a} + 5\mathbf{c} \text{ and } \overrightarrow{AQ} = 4\mathbf{a} + 8\mathbf{c}.$$

$$AP : PC = 4 : 1.$$

(a) Write each of the following in terms of $\mathbf{a}$ and $\mathbf{c}$.

Give your answers in their simplest form.

(i) $\overrightarrow{OP}$.

[2]

(ii) $\overrightarrow{PQ}$.

[2]

(b) (i) Explain why PQ is parallel to CB .

[1]

(ii) Find the ratio $CB : PQ$.

[1]

[N16/I/25]

8. P is the point $(0, 3)$, Q is the point $(4, 11)$ and R is the point $(a, 3)$.

(a) The product $(\text{gradient of } PQ) \times (\text{gradient of } QR) = -1$.

Use this information to show that $a = 20$.

[2]

(b) The line PQ is perpendicular to the line QR .

Use vectors to find the coordinates of the point S , so that $PQRS$ is a rectangle.

[1]

(c) Calculate the area of the rectangle $PQRS$.

[2]

[N15/I/20]

1. (a) P is the point $(-3, 5)$ and Q is the point $(2, 11)$.

$$\overrightarrow{PR} = \begin{pmatrix} 8 \\ -2 \end{pmatrix}.$$

(i) Calculate the length of the line PQ .

[2]

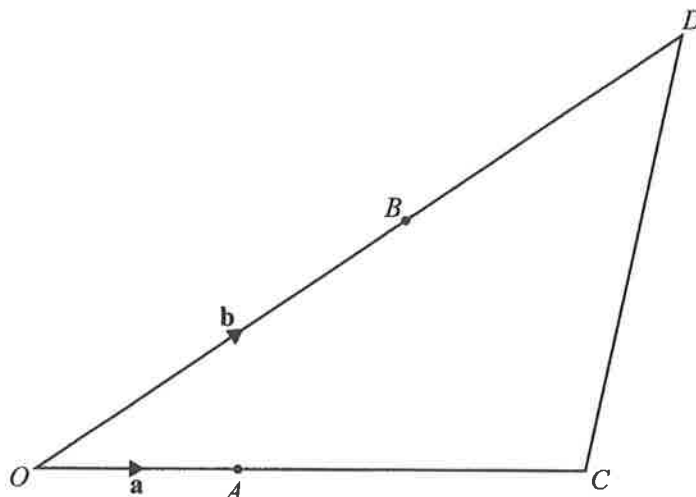
(ii) Find the coordinates of point R .

[1]

(iii) Find the equation of the line QR .

[3]

(b)



OCD is a triangle.

A is a point on OC and B is a point on OD .

$\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

$OA = \frac{1}{3}OC$ and $OB : BD = 3 : 2$.

X is a point on OC such that BX is parallel to DC .

Find $\overrightarrow{XD}$.

Give your answer as simply as possible in terms of $\mathbf{a}$ and $\mathbf{b}$.

[5]

[N22/II/8]

2. (a) The position vector of point R is $\begin{pmatrix} -4 \\ 5 \end{pmatrix}$.

The position vector of point S is $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$.

(i) (a) Find the vector that represents the translation from R to S .

[1]

(b) Find the magnitude of $\overrightarrow{RS}$.

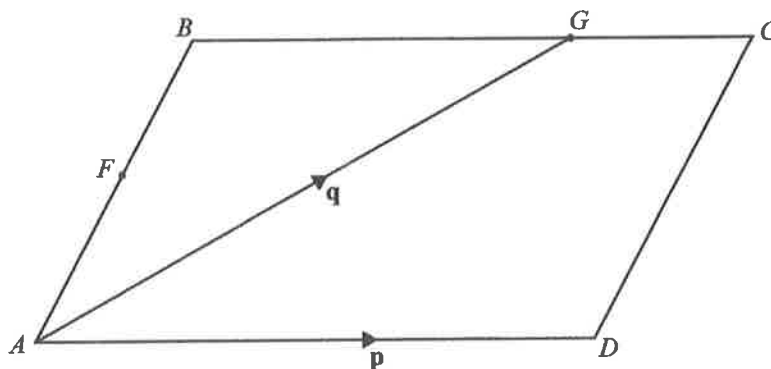
[2]

(ii) The position vector of T is $\begin{pmatrix} k \\ 7 \end{pmatrix}$ and $|\overrightarrow{ST}| = 10$.

Find the two possible values of k .

[3]

(b)



$ABCD$ is a parallelogram.

F is the point on AB such that $AF : AB = 1 : 2$.

G is the point on BC such that $BG : GC = 3 : 1$.

$\overrightarrow{AD} = \mathbf{p}$ and $\overrightarrow{AG} = \mathbf{q}$.

(i) Express $\overrightarrow{CD}$, as simply as possible, in terms of $\mathbf{p}$ and $\mathbf{q}$.

[2]

(ii) H is the point such that $\overrightarrow{DH} = \overrightarrow{GF}$.

Express $\overrightarrow{AH}$, as simply as possible, in terms of $\mathbf{p}$ and $\mathbf{q}$.

[4]

[N21/II/8]

3. $ABCD$ is a parallelogram.

A is the point $(-7, -1)$ and B is the point $(-4, 5)$.

$$\overrightarrow{BC} = \begin{pmatrix} 8 \\ -2 \end{pmatrix}.$$

(a) Find the length of the line AB .

[2]

(b) Find the equation of the line CD .

[4]

(c) X is the point where the diagonals of the parallelogram intersect.

(i) Find $\overrightarrow{XC}$.

[2]

(ii) Find the position vector of X .

[1]

[N20/II/8]

4. (a) P is the point $(4, -3)$, Q is the point $(-5, 1)$.

(i) Write down the column vector $\overrightarrow{PQ}$.

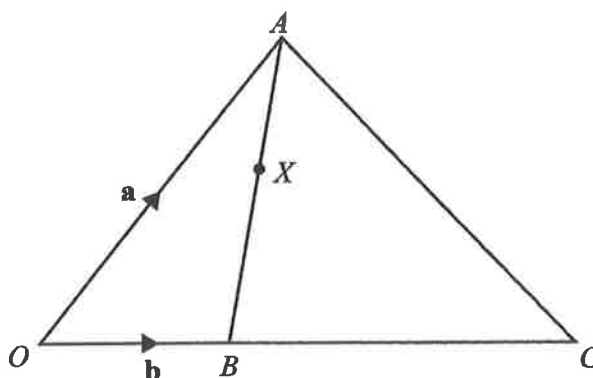
[1]

(ii) R has coordinates $(h, 3)$ and $\overrightarrow{PR} = k\overrightarrow{PQ}$.

Find the value of h and the value of k .

[2]

(b)



OAC is a triangle and B is a point on OC .

$\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $OB : BC = 2 : 3$.

X is the point on AB such that $AX : XB = 1 : 2$.

- (i) Express $\overrightarrow{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$, as simply as possible. [2]
 (ii) Express $\overrightarrow{XB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$, as simply as possible. [2]
 (iii) Y is the point on OC such that $AXYC$ is a trapezium.
 Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, $\overrightarrow{XY}$. [2]

[N19/II/7]

5. $ABCD$ is a parallelogram.

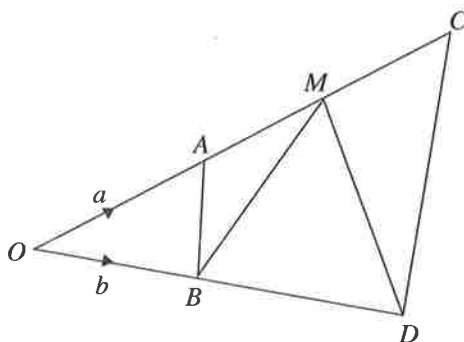
$$\overrightarrow{AB} = \begin{pmatrix} 6 \\ 2 \end{pmatrix} \text{ and } \overrightarrow{BC} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}.$$

The coordinates of vertex C are $(3, -1)$.

- (a) Find the position vector of vertex B . [1]
 (b) Find the coordinates of vertex A and vertex D . [2]
 (c) Calculate angle BAD . [3]
 (d) Calculate the length of diagonal BD . [3]

[N18/II/5]

6.



OCD is a triangle where A and M are points on OC and B is a point on OD .

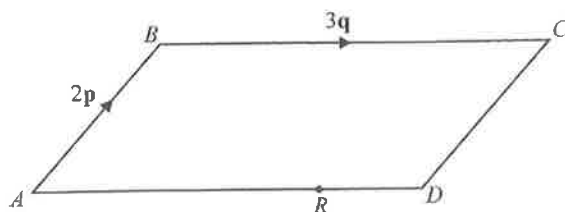
$$\overrightarrow{OA} = \mathbf{a}, \overrightarrow{OB} = \mathbf{b}, \overrightarrow{MB} = \mathbf{b} - 2\mathbf{a} \text{ and } \overrightarrow{MD} = 3\mathbf{b} - 2\mathbf{a}.$$

$$AM = MC.$$

- (a) Express, as simply as possible, in terms of $\mathbf{a}$ and/or $\mathbf{b}$,
 (i) $\overrightarrow{AB}$, [1]
 (ii) $\overrightarrow{BD}$. [1]
 (b) Show that $\overrightarrow{CD} = 3\mathbf{b} - 3\mathbf{a}$. [1]
 (c) Show that triangles OAB and OCD are similar.
 Give a reason for each statement you make. [3]
 (d) Find the ratio of the area of triangle OAB to the area of quadrilateral $ABDC$. [2]

[N15/II/2]

7.



$ABCD$ is a parallelogram.

$\overrightarrow{AB} = 2\mathbf{p}$ and $\overrightarrow{BC} = 3\mathbf{q}$.

R is the point on AD such that $AR : RD = 2 : 1$.

(a) Express $\overrightarrow{RC}$, as simply as possible, in terms of $\mathbf{p}$ and $\mathbf{q}$. [2]

(b) S is a point on AD and T is a point on CD such that $\overrightarrow{RC} = 2\overrightarrow{ST}$.

(i) Show that triangles RCD and STD are similar.

Give a reason for each statement you make. [3]

(ii) Find the ratio area triangle STD : area parallelogram $ABCD$. [2]

(iii) Q is a point on BC .

Triangle QAB is congruent to triangle RCD .

Find $\overrightarrow{QR}$, as simply as possible, in terms of $\mathbf{p}$ and $\mathbf{q}$. [2]

[N17/II/6]